MIL-S-322C
5 February 1968
Superseding
MIL-S-322B
20 September 1962

MILITARY SPECIFICATION

SODIUM NITRATE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers sodium nitrate for use in the manufacture of pyrotechnics, and other ordnance applications (see 6.1).

* 1.2 Classification - Sodium nitrate shall be of the following grades and classes as specified (see 6.2):

Grade A (see Table I)	Class 1 (see Table II)
Grade B (see Table I)	Class 2 (see Table II)
Grade C (see Table I)	Class 3 (see Table II)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

RR-S-366	Sieve, Test
PPP-D-705	Drum: Metal Shipping, Steel (Over 12 and Under 55 Gallon).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6810

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MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1233	Procedures for Determining Particle Size, Particle Size Distribution and Packed Density of Powdered Materials.

(When requesting any of the applicable documents, refer to both title and number. Copies of the applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications - The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

American Society For Testing And Materials Standards

D1193	Reagent Water

(Copies of ASTM Standards may be obtained from the American Society for Testing and Materials 1916 Race Street, Philadelphia, Pa. 19103.)

American Chemical Society Publications
Reagent Chemicals, A.C.S. Specifications

(Copies of A.C.S publications may be obtained from the American Chemical Society, 1155 Sixteenth Street, N.W. Washington, 6, D.C.)

Technical Society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.

Interstate Commerce Commission

49 CFR 71-90	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations and are available from the Superintendent of Documents, Government Printing Office, Washington, 25. D. C. (Orders for the above publications should cite "the latest issue and supplements thereto").

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3. REQUIREMENTS

3.1 Material - The sodium nitrate shall be a product of high quality, suitable for the purpose intended, and so formulated as to meet the requirements specified herein for the applicable grade and class of this specification. (see 4.4.1).

* 3.2 Chemical requirements - The sodium nitrate shall conform to the applicable chemical requirements specified in Table I when tested as specified in the corresponding test paragraphs in Section 4.

TABLE I - Chemical requirements

Property	Grade A		Grade B 1/		Grade C	
	Percentage		Percentage		Percentage	
	Max.	Min.	Max.	Min.	Max.	Min.
Moisture	0.75	...	0.5	...	0.75	...
Insoluble Matter	0.5	...	0.1	...	0.5	...
Alkalinity(as Na_2O)	0.05	...	None	...	0.06	...
Nitrates(as $NaNO_3$)	...	97.0	...	99.5	...	97.0
Chlorates(as $KClO_3$)	0.06	...	None	...	0.06	...
Calcium (as CaO)	0.3	...	0.1	...	0.3	...
Magnesium(as MgO)	0.15	...	0.06	...	0.15	...
Sulfates(as Na_2SO_4)	0.5	...	0.2	...	0.45	...
Chlorides(as NaCl)	...	...	0.15	...	0.15	...

1/ The percentages indicated, except the percent of moisture, are to be obtained in the sodium nitrate after the sample has been dried to constant weight at 302± 5°F(150 ± 3°C).

* 3.3 Granulation - The sodium nitrate shall conform to the granulation requirements of Table II, when determined in accordance with 4.4.11, unless otherwise specified in the contract or order.

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TABLE II-Granulation requirements

Sieve number 1/	Class I	Class 2	Class 3
	Percent through	Percent through	Percent retained on
	Minimum	Minimum	Minimum
20	...	...	80.0
60	98.0	...	...
100	...	98.0	...
200	...	...	...
325	...	...	...

1/ Use U. S. Standard sieves conforming to Specification RR-S-366.

3.4 Average particle diameter - (applicable to Class 2 only) In addition to conforming to the granulation requirements specified in Table II, the average particle diameter of Class 2 sodium nitrate shall be 30± 15 microns, when determined in accordance with 4.4.12.

3.5 Workmanship - The sodium nitrate shall be processed in a manner that will produce the high quality product necessary to meet the requirements of this specification (see 4.4.1).

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection - The examination and testing of the sodium nitrate shall be entirely by quality conformance inspection.

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* 4.3 Quality conformance inspection - The quality conformance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, an examination of filled containers for conformance to packaging, packing, and marking requirements, and examinations and tests of samples (4.3.3) for all the requirements of this specification. The inspection terms used herein are defined in MIL-STD-109.

* 4.3.1 Quality conformance samples - Quality conformance samples shall consist of a sample for tests (4.3.3) and samples for examination of filled containers (4.3.4). Samples shall be labeled completely with information identifying the purpose of the sample, name of product, specification number, lot and batch number, date of sampling, and contract number.

* 4.3.2 Lot and batch - For the purpose of sampling and inspection, a lot shall consist of sodium nitrate from one batch offered for delivery at one time, but not to exceed 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, providing the operation is continuous. A batch is that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform. In the event the process is a batch operation, each batch shall constitute a lot.

* 4.3.3 Sample for tests - Ten (10) primary samples of equal weight, 25 grams each, shall be randomly selected from each lot of material and thoroughly mixed to form a composite sample representative of the lot. The container shall be labeled to identify the sample with the lot represented. If there are less than 10 containers in a lot, a sample shall be taken from each container. The composite sample shall be subjected to the conformance inspection specified herein. A lot shall be unacceptable if a sample fails to meet any of the test requirements specified.

4.3.4 Sample for examination of filled containers - A random sample of filled containers shall be selected from each lot of sodium nitrate in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective.

4.4 Inspection methods - Unless otherwise specified, the physical and chemical values specified in Section 3 shall apply to the average of the determinations made on the unit of product.

4.4.1 Conformance of the sodium nitrate to the requirements for material (3.1) and workmanship (3.5) shall be determined by appropriate examination and testing in accordance with Section 3.

4.4.1.1 Reagent water conforming to D1193 and reagent chemicals conforming to American Chemical Society (A.C.S.) specifications (see 2.2) shall be used throughout the tests. Weighings shall be made to the nearest 0.0001 gram (gm). Unless otherwise specified herein, solutions shall be made up on a weight basis.

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4.4.2 Moisture - Transfer a weighed 10 gm portion of the sodium nitrate to a previously tared shallow weighing bottle, heat in an air bath for 2 hours at 302 ± 5°F. (150 ± 3°C.), cover, cool in a desiccator and weigh. Calculate the loss in weight as the percentage of moisture in the sample.

$$\text{Percent moisture} = \frac{100\ (A-B)}{W}$$

Where:

A = Weight of bottle plus sample before drying in grams.

B = Weight of bottle plus sample after drying in grams.

W = Weight of sample in grams.

4.4.3 Insoluble matter - Transfer a weighed 25 gm portion of sodium nitrate to a beaker. Heat to boiling with 200 ml of water and decant the solution through a filtering crucible of fine porosity (a nominal maximum pore diameter of 4.5 to 5.0 microns) which has been dried at 212 ± 5°F. (100 ± 3°C.) and weighed. Rinse the beaker with boiling distilled water and transfer any adhering insoluble matter to the filter. Decant the wash waters from the beaker through the filter. Catch the filtrate and wash waters in a 500-ml volumetric flask and retain for the determination of alkalinity (see 4.4.4). The filtering crucible shall be dried in an oven at 221 ± 9°F. (105 ± 5°C.) for 3 hours, cooled in a desiccator, and weighed. The increase in weight shall be calculated to percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A-B)}{W}$$

Where:

A = Weight of crucible plus residue in grams.

B = Weight of crucible in grams.

W - Weight of sample in grams.

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4.4.4 Alkalinity - Using freshly boiled distilled water, make up the filtrate obtained as specified in 4.4.3 to 500-ml and mix thoroughly. Pipet 100 ml of this solution into a flask, add 3 to 5 drops of phenolphthalein indicator and if the solution is alkaline, titrate with 0.1N acid. A blank determination shall be made on the distilled water and a correction applied if necessary. Calculate to percentage Na_2O in the sample as follows:

$$\text{Percent } Na_2O = \frac{3.1\ VN}{W}$$

Where:

V = ml acid used in the titration.

N = Normality of acid used.

W = Weight in grams of sample represented by the aliquot.

4.4.5 Nitrates

4.4.5.1 Standardization of the nitrometer

4.4.5.1.1 Absolute method - Assemble the nitrometer (DuPont or equivalent) after thoroughly cleaning and drying all parts. The various parts, with the exception of the reservoir bulbs, are completely filled with mercury. Place 25 ml of 94.5 ± 0.5 percent sulfuric acid in the generating bulb. Draw air into the bulb to nearly the level of the lower shoulder by lowering the mercury reservoir and opening both stopcocks. Close both stopcocks and shake vigorously for 3 minutes. Connect the generating bulb into the compensating tube by adjustment of the two mercury reservoirs, keeping open the stopcock of the measuring tube. Continue the introduction of the air until the mercury in the reading tube is nearly at the 12.50 mark. Seal the compensating tube by means of a small blowpipe flame. Desiccate similarly in the generating bulb another portion of air and transfer this to the reading tube until the mercury in the reading tube, compensating tube, and their mercury reservoirs are at the same level, the mercury in the reading tube being approximately at the 12.50 mark. It is not necessary that these gas volumes be exact or equal. Fill a small U-tube manometer with sulfuric acid and attach to the reading tube. Allow the apparatus to come to constant temperature and then adjust the mercury levels in the reading tube, compensating tube, and their mercury reservoirs so that the air in the reading tube is under atmospheric pressure, as shown by the acid levels in the arms of the manometer tube. Take the reading on the graduated scale of the calibrated reading tube, and at the same time note the temperature and barometric pressure. From these readings calculate the volume of air at 20°C. and 760 mm. pressure in the reading tube by means of the following formula:

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carboxy terminated polybutadiene liquid polymer containing a nominal

$$V = R \frac{P \times 293(1-0.00018t)}{760(273 + T)(1-0.00018 \times 20)} = \frac{0.38692\ PR(1.0.00018t)}{273 + T}$$

Where:

V = ml air at 20°C. and 760 mm., barometer at 20°C.

R = ml air at the observed temperature (T°C.) and at the observed pressure (Pmm).

t = Temperature (°C.) of the barometer.

The factor 0.00018 represents the correction to be applied for the coefficient of expansion of mercury in the barometer. When V has been calculated, adjust the heights of the reading tube, compensating tube, and their reservoirs so that, with the mercury in the reading and compensating tubes on a level, the mercury in the reading tube marks the calculated volume of the air. Paste a strip of paper on the tube so as to mark the height of the mercury in the compensating tube. Take this as the standard volume of air with which every volume of gas to be measured is compared.

Note. The use of dry air tends to form a scum around the reading level of the compensating tube, thus requiring frequent restandardization of the nitrometer. This scum is caused by reaction of oxygen in the air with mercury or by impurities. Dry nitrogen eliminates this condition and should be used, if available. A half drop of sulfuric acid will also prevent this scum.

4.4.5.1.2 Potassium nitrate method - Recrystallize three times from 95 percent alcohol a sample of chemically pure grade of potassium nitrate, and grind this to pass through a U. S. Standard No. 100 sieve. After drying at 275 to 302°F. (135 to 150°C.) for 2 or 3 hours, weigh a portion of exactly 1 gm into a small weighing bottle and transfer this portion to the cup of the generating bulb. Wash the potassium nitrate into the generating bulb by means of 8 to 10 portions of 94.5 + 0.5 percent nitrogen-free sulfuric acid cooled to approximately 41°F. (5°C.) using a total volume of 25 ml of the acid. With the lower stopcock of the generating bulb open and the mercury reservoir low enough to maintain a slightly reduced pressure in the generating bulb, shake the sample until most of the gas has been generated. Adjust the mercury reservoir until the mercury drops nearly to the lower shoulder of the generating bulb. Close the lower stopcock and shake the bulb vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and allow the tube to stand for 5 minutes, the levels in the generating bulb and reservoir being approximately the same. 1/ Close the lower stopcock, and shake the generating bulb for 1 additional minute. Transfer the gas to the reading

1/ Having previously introduced desiccated air into the compensating tube as described under the absolute method.

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tube, and adjust the levels of the mercury in the reading and compensating tubes to approximately the same height. Allow the gas to stand for about 20 minutes to permit equalization of the temperature. With a sulfuric acid U-tube leveling device, make careful adjustment, so that the mercury in both tubes are at the same level and that in the reading tube it is at the 13.85 mark (the percentage of nitrogen in potassium nitrate). Make a mark in the compensating tube to show the volume occupied by the air equivalent to the volume of nitric oxide gas liberated from exactly 1 gm of potassium nitrate. This mark is usually made on a strip of gummed paper attached to the compensating tube.

Note 1: In order to avoid any doubt regarding the purity of the potassium nitrate, it is desirable to check the standardization obtained by the use of potassium nitrate by means of the absolute method. If the agreement is within one-or two-hundreths of 1 percent, the potassium nitrate can be considered as of satisfactory quality and used in subsequent periodic standardizations. It has been found by actual experimentation that the solubility of nitric oxide in sulfuric acid of the strength prescribed is so small that no correction need be applied to bring the value in agreement with the mark set by the absolute method.

Note 2: If the nitrometer is standardized by both the absolute and potassium nitrate methods, the results may not be in agreement by 0.01 to 0.04 percent nitrogen due to expansion of the brass barometer scale. Such difference may be removed by applying the brass scale correction corresponding to the temperature.

4.4.5.2 Procedure - Crush a portion of about 3 gm of the sodium nitrate so that it will pass through a U. S. Standard No. 100 sieve and dry for 2 hours at 302 ± 5°F. (150 ± 3°C.). Accurately weigh approximately 0.85 gm of the dried sample and transfer it to the cup of a standardized five-part nitrometer (DuPont or equivalent). Measure into a small graduate 25 ml of nitrogen-free 94.5 ± 0.5 percent sulfuric acid cooled to approximately 41°F. (5°C.). Wash out the weighing bottle into the nitrometer cup with 5 ml of the sulfuric acid. Lower the mercury reservoir sufficiently to maintain a reduced pressure in the generating bulb. Open wide the upper stopcock and open the lower stopcock sufficiently to draw the acid mixture into the generating bulb. Make successive 4 ml rinsings with the rest of the sulfuric acid until it has all been drawn in. Close the upper stopcock, leaving the lower one open, and adjust the mercury reservoir just low enough to maintain a slightly reduced pressure in the generating bulb. Shake the bulb gently until most of the gas has been generated. Lower the mercury reservoir until the level of the mercury in the generating bulb is at the height of the lower shoulder, then close the lower stopcock and shake vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and adjust the mercury reservoir until the mercury in the generating bulb is at approximately the same height as the mercury in the reservoir. This will bring the pressure inside the generating bulb to approximately one atmosphere, and the solubility of the nitric oxide gas in the sulfuric acid will be normal. Close the lower stopcock and shake the bulb vigorously for an additional 1 minute.

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Transfer the gas to the measuring tube, adjust the levels of the mercury in the compensating and measuring tubes to approximately the same height. Let the gas stand for 20 minutes in order to permit equalization of the temperature of the gas in the two tubes, and adjust the mercury levels closely with a sulfuric acid U-tube leveling device, the mercury level in the compensating tube being at the standard volume mark. Calculate the percentage of sodium nitrate as follows:

$$\text{Percent sodium nitrate} = \frac{6.07R}{W}$$

Where:

R = Reading of the measuring tube.

W = Weight of dry sample in grams.

* 4.4.5.3 Alternate method: The DEVARDAS method consists of placing the sample in a Kjedahl flask, adding water, caustic soda and DEVARDAS reducing alloy to the flask and heating the mixture to reduce the nitrates to ammonia. The ammonia is distilled over and the excess hydrochloric acid is titrated with .IN Sodium hydroxide. The purity of the nitrate is calculated from results obtained.

* 4.4.5.3.1 Procedure. Using a 250 ml Erlenmeyer flask add 35 ml of standard acid (.IN hydrochloric) with a few drops of methyl red indicator. Connect the flask to the outlet end of a condenser so that the condenser tip is immersed in the liquid. To a Kjeldahl flask, add consecutively 200 to 250 mg of 1 gm of the sample, Devardas Alloy, 100 ml water, and 5 ml of 20 % sodium hydroxide. Quickly wash down the walls of the flask with 25 ml water and connect it to the condenser through a spray trap. Be sure all connections are gas tight before proceeding further. Heat the flask to boiling and continue the distillation until 100 ml of distillate has been collected. Lower the receiving flask until the tip of the condenser is out of the distillate and stop the distillation. Disconnect the spray and wash the inside of the condenser into the receiver with water. Titrate the contents of the receiver with .IN sodium hydroxide.

CAUTION: Too much alloy may cause excessive foaming and consequent contamination of the distillate.

Calculate the percentage purity of sodium nitrate as follows:

$$\text{Percent Purity} = \frac{(VaNa-VbNb)F}{10nw}$$

Where:

Va = Titer or aliquot of standard acid in ml.

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Na = Normality of the standard acid.

Vb = Titer or aliquot of standard base in ml.

Nb = Normality of the standard base.

n = Number of nitrogen atoms in the formula.

w = Weight of sample in grams corrected for moisture.

F = Formula weight of the nitrate being determined.

4.4.6 Chlorates - Transfer a weighed 25 gm portion of sodium nitrate to a beaker and dissolve in hot distilled water. Add 5 ml of HNO_3 and an excess of silver nitrate solution and stir. Filter the solution to remove any chlorides which may have been precipitated. Add to the solution 10 ml of 30 percent formaldehyde solution and heat on the water bath for 1/2 hour. Chlorate is thus reduced to chloride and precipitated by the excess silver nitrate. Filter off the precipitate on a tared filtering crucible which has previously been dried at 212 ± 5°F. (100 ± 3°C.) and which has a nominal maximum pore diameter of 4.5 to 5.0 microns. Dry the crucible and precipitate for 3 hours at 212 ± 5°F. (100 ± 3°C.). Cool in a desiccator and weigh. Calculate to percentage of potassium chlorate as follows:

$$\text{Percent potassium chlorate} = \frac{85.5A}{W}$$

Where:

A = Weight of the precipitate in grams.

W = Weight of sample in grams.

* 4.4.7 Calcium - By means of a pipet, transfer to a beaker 100 ml of the solution prepared as specified in 4.4.4. Make distinctly acid with 5 ml of concentrated HNO_3. Then make disticntly alkaline with NH_4OH and heat to boiling. Precipitate the calcium by adding 10 ml of a saturated solution of ammonium oxalate. Continue the boiling for a few minutes, filter the precipitate on #42 Whatman paper or equivalent, wash with cold distilled water containing a little NH_4OH, and determine the percentage of calcium oxide either gravimetrically or volumetrically. (Save the filtrate for the magnesium determination in 4.4.8).

* 4.4.7.1 Gravimetric determination. Place the precipitate and filter paper in a weighed crucible and cover. Heat gently until the water is expelled and the paper charred. Transfer the crucible and contents to a muffle furnace and heat to 1200°C for 5 to 10 minutes. Remove the cover for an instant from time to time to assist the escape of carbon dioxide during the heating. Cool the covered crucible and its contents in a desiccator containing concentrated sulfuric acid. Weigh the cooled calcium oxide and repeat ignition until the weight is constant. Calculate the percent of calcium oxide as follows:

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$$\text{Percent calcium oxide} = \frac{100(A-B)}{W}$$

Where: A = Weight of crucible plus residue in grams.
B = Weight of crucible in grams.
W = Weight in grams of sample represented by the aliquot.

4.4.7.2 Volumetric determination - Wash the filter paper and precipitate thoroughly and then transfer paper and precipitate to a beaker containing 400 ml hot water to which has been added 10 ml of 18N sulfuric acid. Disintegrate the paper by stirring vigorously with a glass rod and then titrate rapidly while hot with approximately 0.05N $KMnO_4$ solution until a pink end point persists for one-half minute.

Calculate the percent of calcium oxide as follows:

$$\text{Percent calcium oxide} = \frac{2.8\ VN}{W}$$

Where:

V = ml of $KMnO_4$ solution used.

N = Normality of $KMnO_4$ solution.

W = Weight in grams of sample represented by the aliquot.

This method is preferred for small amounts of calcium since it is more accurate.

4.4.8 Magnesium - Catch the filtrate and wash water from the calcium determination (see 4.4.7) in a beaker. Add 10 ml of a 10 percent solution of disodium phosphate and then a volume of concentrated ammonium hydroxide (27 to 30 percent) of 1/10 volume of the beaker. Mix thoroughly and allow to stand overnight. Filter the precipitate on #42 Whatman paper or equivalent, wash with 5 percent NH_4OH and ignite in a tared platinum crucible. If the ignition does not result in a white precipitate, cool the crucible and contents, add a few drops of nitric acid or ammonium nitrate solution, heat gradually and then ignite fully. Spattering should be avoided especially during the early part of the heating. Cool in a desiccator and weigh. Calculate the increase in weight to percent magnesium oxide as follows:

$$\text{Percent magnesium oxide} = \frac{36.2(A-B)}{W}$$

Where:

A = Weight of precipitate plus crucible in grams.

B = Weight of crucible in grams.

W = Weight in grams of sample represented by the aliquot.

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4.4.9 Sulfates - By means of a pipet, transfer to a beaker 200 ml of the solution prepared as specified in 4.4.4. Add sufficient HCl to make the solution slightly acid, and heat to boiling. Slowly add 10 ml of a 10 percent solution of $BaCl_2$ and continue boiling for several minutes. Settle the precipitate on a steam bath and then filter in a tared filtering crucible of fine porosity, having a nominal maximum pore diameter of 4.5 to 5.0 microns. Wash with hot distilled water. Ignite, cool in a desiccator and weigh. Calculate the increase in weight to percentage of sodium sulfate as follows:

$$\text{Percent sodium sulfate} = \frac{60.9(A-B)}{W}$$

Where:

A = Weight of precipitate plus crucible in grams.

B = Weight of crucible in grams.

W = Weight in grams of sample represented by the aliquot.

4.4.10 Chlorides - An accurately weighed 25 gm portion of sodium nitrate shall be transferred to a 600 ml beaker. The sample shall be dissolved in 300 ml of hot approximately 185°F. (85°C.) distilled water and 5 ml of concentrated nitric acid added. Twenty-five milliliters of approximately 0.1N silver nitrate solution shall be added to the solution with the aid of a pipet. The solution shall be heated to boiling on a hot plate with occasional stirring and boiling moderately for approximately 2 minutes with constant stirring. The beaker and contents shall be removed from the hot plate. The precipitate shall be allowed to settle (if the precipitate does not settle readily it is permissible to filter the solution as specified and test the filtrate) and one drop of the supernatant liquid shall be added to one drop of a one percent solution of sodium chloride on a black porcelain spot plate. If no precipitate occurs on the spot plate, a second 25 ml portion of silver nitrate solution shall be added to the sample, boiled as indicated above, and the spot test repeated. The addition of silver nitrate solution shall be continued and the spot test repeated as specified above until a precipitate of silver chloride is obtained. The supernatant solution shall be filtered through a tared pyrex sintered glass crucible having a nominal maximum pore diameter of 4.5 to 5.0 microns, and with the aid of suction, the filtrate shall be collected in a 1000 ml filtering flask or 600 ml beaker. The silver chloride precipitate shall be transferred quantitatively to the filtering crucible with the aid of five 20 ml portions of 1:100 nitric acid solution and one 20 ml portion of distilled water. The crucible and contents shall be dried in an oven at 230 ± 9°F. (110 ± 5°C.) for 3 hours. The crucible shall be cooled in a desiccator and weighed. The percent chlorides, as sodium chloride, in the sample shall be calculated as follows:

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$$\text{Percent sodium chloride} = \frac{40.8(A-B)}{W}$$

Where:

A = Weight of crucible plus precipitate in grams.

B = Weight of crucible in grams.

W = Weight of sample in grams.

4.4.11 Granulation - Attach a bottom pan to the required U. S. Standard sieves. Place two metal washers (flat SAE standard galvanized, OD 1 1/16 in., ID 17/32 in., thickness 0.094 in.) and a 100 gm portion of the sample on the sieve, cover and shake for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts retained or passing through the sieves and calculate the percentage as required.

4.4.12 Average particle diameter (applicable to Class 2 only)- Determine the average particle diameter by the procedure specified in MIL-STD-1233, Method 100.

* 4.4.13 Examination of filled containers - Each sample of filled containers selected in accordance with 4.3.4 shall be examined for defects of construction of the container and closure, evidence of leakage, and unsatisfactory markings. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Not applicable

5.2 Packing - Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A - In addition to the following, packing shall be in accordance with the Code of Federal Regulations "49 CFR 71-90".

5.2.1.1 Exterior containers - Sodium nitrate shall be packed in metal drums in accordance with PPP-D-705, Type III. Gross weight of container shall not exceed 120 lbs. Containers shall be provided with 0.004 inch polyethylene liners, sealed by heat sealing.

5.2.2 Level C - Sodium nitrate packed in accordance with the Code of Federal Regulations "49 CFR 71-90" shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall conform to the carrier rules and regulations applicable to the mode of transportation.

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5.3 Marking - Unless otherwise specified in the contract or order, marking shall be in accordance with Standard MIL-STD-129 and Code of Federal Regulation "49 CFR 71-90."

6. NOTES

6.1 Intended use - The sodium nitrate covered by this specification is intended for use in the manufacture of pyrotechnic compositions and incendiary mixtures.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Grade and class of material (see 1.2).
(c) Quantity required in pounds.
(d) Type and capacity of containers in which the sodium nitrate is to be furnished, if other than as specified in Section 5.
(e) Applicable levels of packing, and marking with requirements in detail, if other than as specified in Section 5.
(f) Instructions regarding availability of inspection records.

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Air Force - 68

* Review Activities:

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Navy - OS, AS
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* User activity:

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Preparing activity:

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